



Quantum **ROI** Business

Business Optimization Intelligence

Turn complex business problems into measurable optimization opportunities.



OPTIMIZE
Operations



ANALYZE
Scenarios



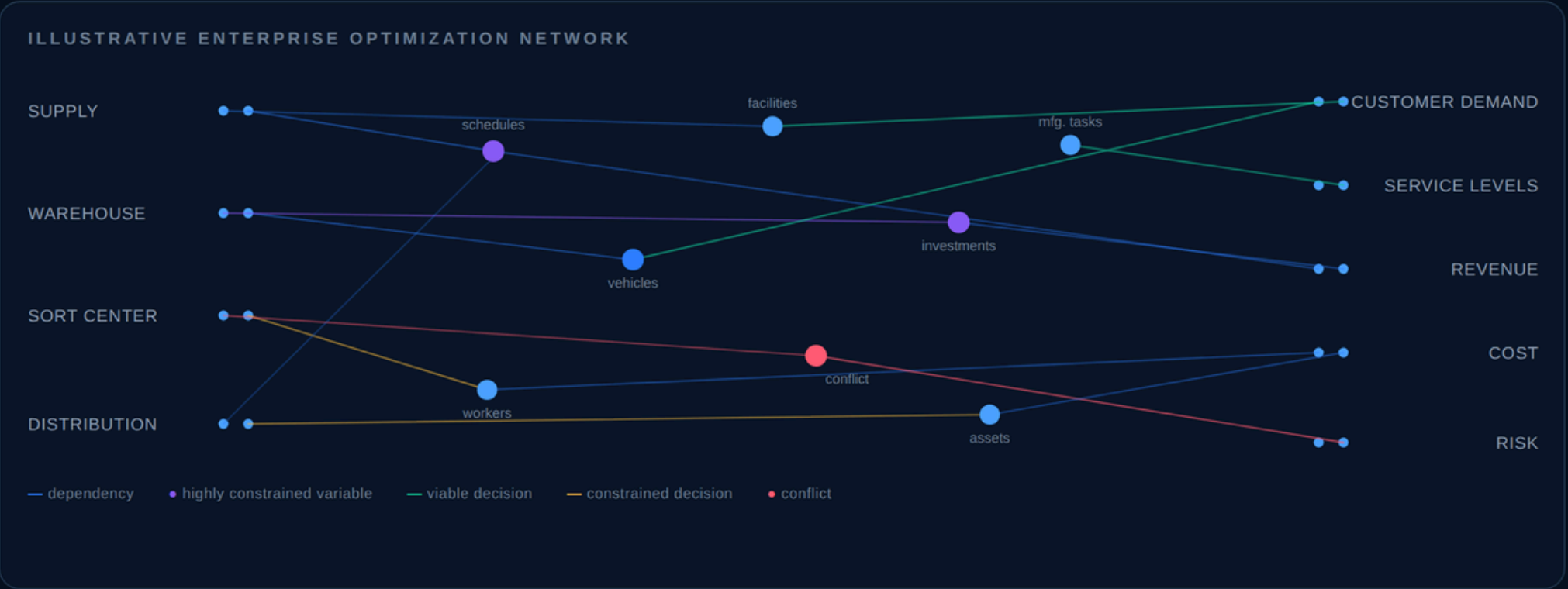
QUANTIFY
Business Value



EXPLORE
Quantum Readiness

BUSINESS DECISIONS ARE BECOMING OPTIMIZATION PROBLEMS.

Cost, capacity, timing, resources, risk, and service constraints interact — making seemingly simple operational decisions mathematically difficult.



TOO MANY INTERDEPENDENT DECISIONS

- › Thousands of possible combinations
- › Multiple constraints
- › Conflicting objectives



THE BEST-LOOKING PLAN MAY NOT BE FEASIBLE

- › Capacity · Timing · Coverage
- › Resources · Dependencies · Risk



BUSINESS VALUE IS HARD TO QUANTIFY

- › What changes? How much?
- › Under which assumptions?
- › How sensitive is the result?



THE CHALLENGE IS NOT FINDING A NUMBER.
IT IS FINDING A **FEASIBLE DECISION**.

- Optimize the objective.
- Respect the constraints.
- Understand the trade-offs.
- Measure the uncertainty.

ILLUSTRATIVE OPTIMIZATION PROBLEM				SYNTHETIC DEMONSTRATION DATA		
ANNUAL COST	NEXT-DAY COVERAGE	AVG ROUTE TIME	CAPACITY UTILIZATION	DECISION COMPLEXITY	VARIABLES	CONSTRAINTS
\$126.8M	71.4%	9.2 hrs	88%		142	37
						DATA RECORDS
						8,420

Complex optimization is a business problem before it is a technology problem.

TURN COMPLEX BUSINESS PROBLEMS INTO OPTIMIZATION OPPORTUNITIES.

Quantum ROI Business creates a structured workflow from business problem definition to optimization evidence and measurable economic impact.

STRUCTURED WORKFLOW

01

DEFINE

Business problem

02

MODEL

Variables + constraints

03

OPTIMIZE

Classical methods

04

EXPLORE

Scenarios + sensitivity

05

FORMULATE

QUBO / Ising

06

EVALUATE

Quantum readiness

07

DECIDE

Business impact

Create Business Case

Run Analysis

Run Simulation

USE CASES — START WITH THE OPERATIONAL PROBLEM, NOT THE TECHNOLOGY

LOGISTICS

Middle-Mile Delivery

Optimize network movement, capacity, timing, and service levels.

142 variables · 37 constraints

READY

WORKFORCE

Workforce Rostering

Build feasible staffing plans while controlling labor cost and coverage.

280 variables · 64 constraints

READY

FINANCE

Portfolio Optimization

Balance return, risk, liquidity, diversification, and decision constraints.

48 variables · 22 constraints

FORMULATED

MANUFACTURING

Robot Path Planning

Optimize task sequencing, resource allocation, robot movement, and throughput.

184 variables · 61 constraints

NEEDS REVIEW

ONE WORKFLOW. MANY OPTIMIZATION PROBLEMS.

The application separates the business objective from the optimization method — making it possible to compare approaches rather than betting everything on one algorithm.

BASELINE FIRST

Every result is compared to a reference plan.

CONSTRAINT AWARE

Feasibility is evaluated alongside the objective.

EVIDENCE BACKED

Assumptions, runs, scenarios, and formulations can be inspected.

QUANTUM ROI BUSINESS

Optimization Intelligence

Monitor active business cases, modeled value, and quantum-readiness.

+ Create Business Case

Explore Use Cases

Illustrative enterprise demo data — synthetic demonstration data

KEY METRICS

Active Opportunities

12

4 require review · +2

Modeled Annual Value

\$42.8M

across 12 scenarios · +\$3.2M

Optimization Runs

184

last 30 days · +18

PRIORITY OPPORTUNITIES

Middle-Mile Delivery · Logistics

Baseline \$126.8M → \$109.7M

Medium

Workforce Rostering · Workforce

Baseline \$31.7M → \$29.9M

High

Portfolio Allocation · Finance

Modeled value \$9.6M

Experiment

FROM BUSINESS PROBLEM TO MATHEMATICAL EXPERIMENT.



The diagram illustrates the structure of a business problem, organized into four main sections: BUSINESS PROBLEM, OBJECTIVE, SUB-OBJECTIVE, and DECISIONS. Below these are the CONSTRAINTS, which are represented by four rounded rectangular buttons.

- BUSINESS PROBLEM** → **STRUCTURED MODEL**
- OBJECTIVE**
 - Minimize network operating cost**
- SUB-OBJECTIVE**
 - Increase next-day service coverage**
- DECISIONS**
 - Vehicle assignments · Facility routing · Capacity allocation · Timing
- CONSTRAINTS**
 - Capacity
 - Service windows
 - Resource availability
 - Coverage

Objective and constraints become structured mathematical inputs — 184 logical variables, 61 constraints.

Classical Benchmark		
Method	Status	Result
Baseline	Reference	\$126.8M
Constraint Programming	COMPLETE	\$111.2M
Random-Key Optimization	COMPLETE	\$109.7M
Simulated Annealing	COMPLETE	\$112.6M
Quantum-Ready Formulation	PREPARED	Pending

COMPARE BEFORE YOU CLAIM.
Illustrative synthetic benchmark · classical methods remain essential



QUANTUM READINESS

SIMULATED / DEMO

Logical variables	184
Reduced kernel	64
QUBO density	18.7%
Connectivity	Sparse
Embedding complexity	Moderate
Classical benchmark	RKO
Cloud experiment	Not submitted

SIMULATION READY

CLOUD EXPERIMENT REQUIRES REVIEW

QUBO MATRIX

$$H(x) = x^T Q x$$

	x_1	x_2	x_3	x_4	x_5
x_1	1	-1	0	0	0
x_2	-1	1	-1	0	0
x_3	0	-1	1	-1	0
x_4	0	0	-1	1	0
x_5	0	0	0	0	1

MEASURE. COMPARE. ITERATE.

- iterate — each cycle re-baselines the decision

BETTER OPTIMIZATION DECISIONS START WITH BETTER EVIDENCE.

Quantum ROI Business makes optimization trade-offs visible — from baseline performance and constraints to scenarios, economic impact, and technical readiness.



QUANTUM ROI BUSINESS

MAKE COMPLEX DECISIONS MEASURABLE.

Model the problem. Compare the options. Understand the trade-offs. Measure the value. Explore the future.

Optimization first.
Evidence always.
Quantum when it makes sense.

QUANTUM

Readiness Lab

From classical to quantum — when it makes sense.

- QUBO Lab
- Scenarios
- Sensitivity

Quantum formulation ready for experimentation

Problem familyMixed integer optimization

Logical variables184

QUBO density18.7%

Reduction184 → 64

FormulationQUBO / Ising

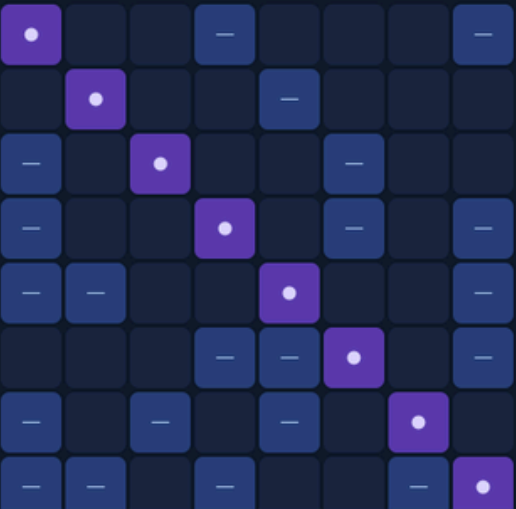
Current benchmarkRKO

Quantum executionNot submitted

QUBO MATRIX VISUALIZATION

$H(x) = x^T Q x$

View equation



Optimization Runs

+ New Run

Experiment Log

Workbench

All

Completed

Running

Needs Review

Failed

Archived

EXP-1084

Completed

Middle-Mile Delivery

SolverRKO Runtime48s
Baseline\$126.8M Optimized\$109.7M

Evidence92%2026-09-24

EXP-1083

Completed

Workforce Rostering

SolverCP-SAT Runtime3m 12s
Baseline\$31.7M Optimized\$29.4M

Evidence87%2026-09-24

EXP-1082

Running

Portfolio Allocation

) Running optimization...

SolverQAOA Runtime—
Baseline8.2% return Optimized9.6% return

EXP-1081

Needs Review

Robot Path Planning

SolverQUBO Runtime1m 8s
Baseline4.2 hrs Optimized3.1 hrs

Evidence78%2026-09-23

EXP-1080

Completed

Vehicle Routing

Optimization Use Cases

Start with the operational problem — not the technology.

Search business problems...

All

Logistics

Workforce

Finance

Manufacturing

Energy

Logistics

Ready

Middle-Mile Delivery

Optimize movement between supply warehouses, sort centers, and distribution stations.

Route costNext-day coverageTravel time+1

VARIABLES142
CONSTRAINTS37
COMPLEXITYHigh

Workforce

Ready

Workforce Rostering

Create optimal shift schedules with skill and coverage constraints across a distributed workforce.

Staffing costCoverageOvertime+1

VARIABLES280
CONSTRAINTS64
COMPLEXITYHigh

Logistics

Formulated

Vehicle Routing

Assign delivery routes to minimize fleet cost while satisfying service windows and capacity.

Fleet utilizationDistanceDriver hours+1

VARIABLES96
CONSTRAINTS28
COMPLEXITYMedium

Finance

Ready

Portfolio Optimization

Reports

Business Impact

Evidence Trail

MODELED ANNUAL VALUE

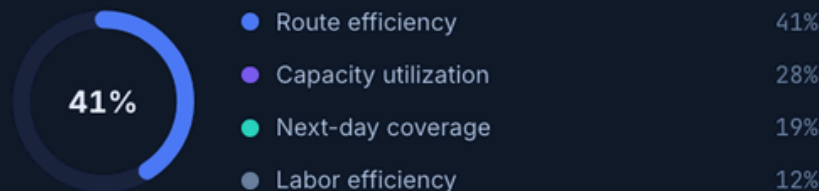
\$18.4M

potential modeled impact — synthetic demonstration data

Cost Reduction\$17.1M
Investment\$3.8M
Net Value\$14.6M

Scenario range\$9.2M — \$23.9M

VALUE DRIVERS



RISK FACTORS

Data quality	● Medium
Integration complexity	● Medium
Adoption	● Medium
Model uncertainty	● High
Quantum availability	● Informational

EXECUTIVE SUMMARY

Opportunity
Improve the operational performance of business processes by leveraging quantum computing.

5. REAL-WORLD IMPACT

From optimization to measurable value.

Quantum ROI Business helps you quantify the business impact of optimization, compare scenarios, assess risks, and decide when quantum experimentation makes sense.



Lower Costs
Reduce operational and capital costs.



Higher Efficiency
Improve resource utilization and throughput.



Better Service
Increase coverage, reliability and customer satisfaction.



De-Risk Decisions
Understand trade-offs, assumptions and uncertainty.



Scenario Comparison ⓘ

Annual value (USD millions)



Scenario	Annual value (USD millions)
Conservative	\$8.0M
Base	\$12.0M
Expansion	\$18.0M

-  Implementation: 9 – 18 months
-  Adoption Rate: 50 – 90%
-  Risk Adjustment: Low – High

Key Insights

- ✓ Middle-Mile Delivery offers the highest modeled value at \$18.4M annually.
- ✓ Optimization reduces operating cost by 13.2% on average.
- ✓ Results are sensitive to adoption rate and implementation cost.
- ✓ Quantum formulation is ready for experimentation (64-variable reduced kernel).

Quantum Readiness READY

Logical variables	184
Reduced kernel	64
QUBO density	18.7%
Connectivity	Sparse
Embedding complexity	Moderate
Current benchmark	RKO

[View Technical Details →](#)

Proven Across Multiple Use Cases

The same platform. Different industries. Real operational challenges.

[Explore All Use Cases →](#)



Logistics & Delivery
Optimize routes, capacity and service levels.



Workforce
Build feasible schedules and control labor costs.



Finance
Balance return, risk, liquidity and diversification.



Manufacturing
Sequence tasks, allocate resources and increase throughput.



Healthcare
Improve enrollment, resource allocation and outcomes.



Energy
Optimize generation, load and reserve margins.



Marketing
Maximize reach, conversion and ROI.



Travel & Mobility
Schedule fleets, crews and gates efficiently.